# CropYield Predictor

## Agricultural Intelligence & Analytics Platform

# Executive Summary

This report documents an end-to-end demonstration pipeline for crop-yield prediction. The included data is synthetic and designed for software demonstration; it is not an official Kaggle dataset or evidence for agricultural decisions.

# Dataset and validation

The demonstration dataset contains 1,800 rows and 14 columns, including categorical crop/region/season/soil/irrigation variables and numeric weather, nutrient, area, and yield fields. Missing values are intentionally present in selected predictors to exercise pipeline imputation. A data dictionary is included.

# Feature Engineering

The pipeline derives total NPK, N:P and K:N ratios, rainfall categories, a humidity-temperature index, and a growing-degree-day proxy. Transformations are wrapped in a scikit-learn Pipeline together with imputation, scaling, one-hot encoding, and the estimator.

# Model Comparison — 5-fold Cross-Validation

| Model | CV MAE | CV RMSE | CV $R^2$ |
|---|---|---|---|
| Random Forest | 0.938 | 2.476 | 0.9918 |
| Gradient Boosting | 1.196 | 2.638 | 0.9909 |
| Extra Trees | 1.013 | 2.747 | 0.9901 |
| Linear Regression | 2.254 | 4.023 | 0.9787 |
| Ridge | 2.250 | 4.032 | 0.9787 |

## Best model holdout metrics

Best cross-validation model: Random Forest. Holdout MAE = 0.760, RMSE = 1.958, $R^2$ = 0.9942. These figures are based only on the synthetic demo dataset.

# EDA Findings (Demonstration Only)

The EDA package includes 19 charts covering yield distributions, mean yield by crop/region/season, rainfall/temperature/humidity/fertilizer relationships, nutrient distributions, yield by soil and irrigation, correlation, feature importance, residuals, and error distribution. Use the figures as a reproducible workflow template; re-interpret them after replacing the demo dataset.

# Recommendations

1. Replace synthetic data with a properly licensed, sourced crop-yield dataset. 2. Confirm units and target definition before modeling. 3. Check for region/year leakage and duplicate observations. 4. Compare CV metrics and residuals after retraining. 5. Treat predictions as exploratory estimates, not agronomic advice. 6. Document data provenance and limitations in the final internship submission.

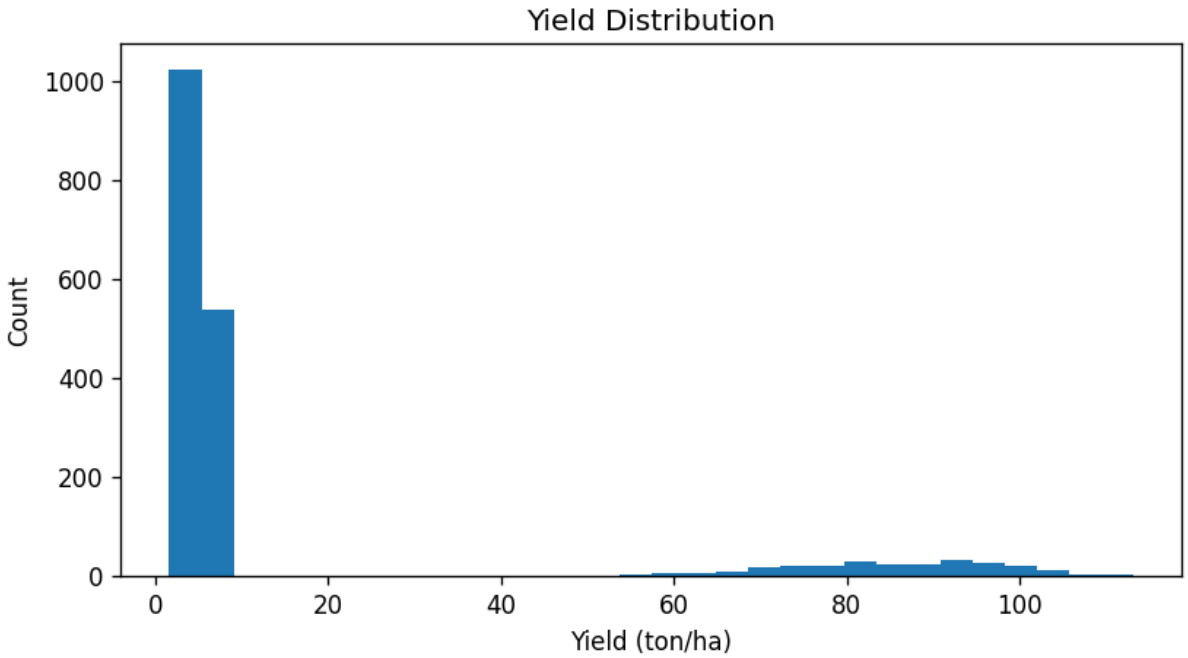
# Dashboard and Reproducibility

The Streamlit app provides EDA Insights, Yield Prediction, and Model Comparison tabs. The trained Pipeline is persisted with Joblib, allowing predictions without retraining. Run `python train.py` to regenerate the comparison and model.

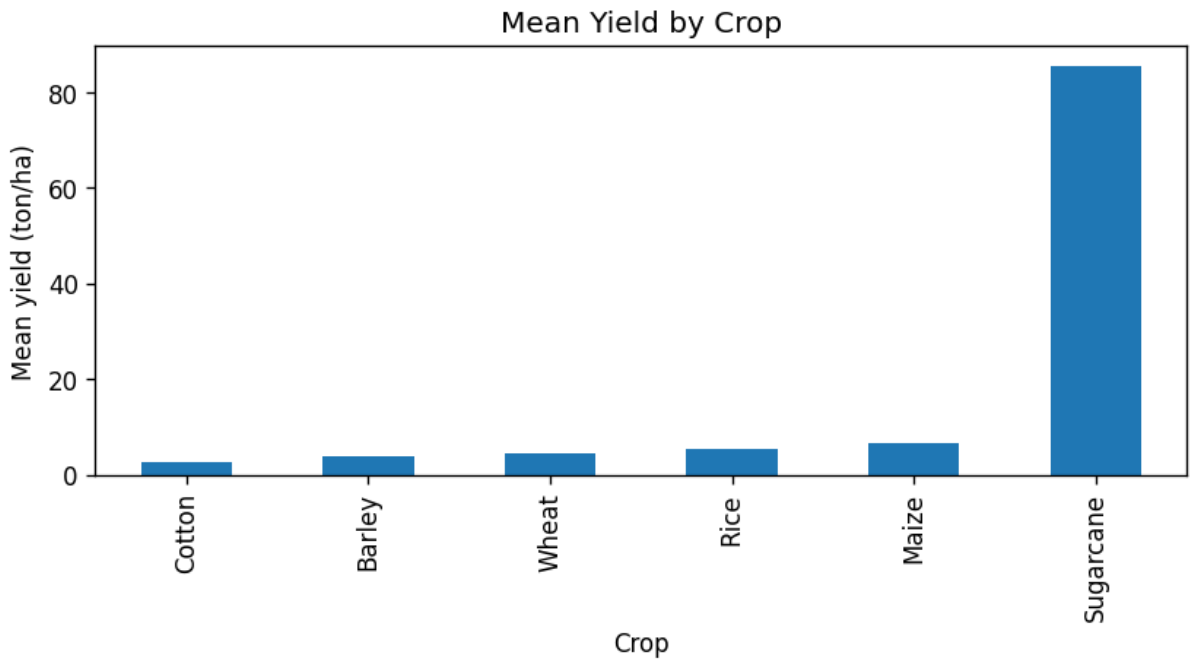
# Limitations

This package includes synthetic data, and its high model scores reflect the generated relationships. They do not establish performance on real-world agricultural data. External dataset schema may differ and should be normalized to the documented columns before training.

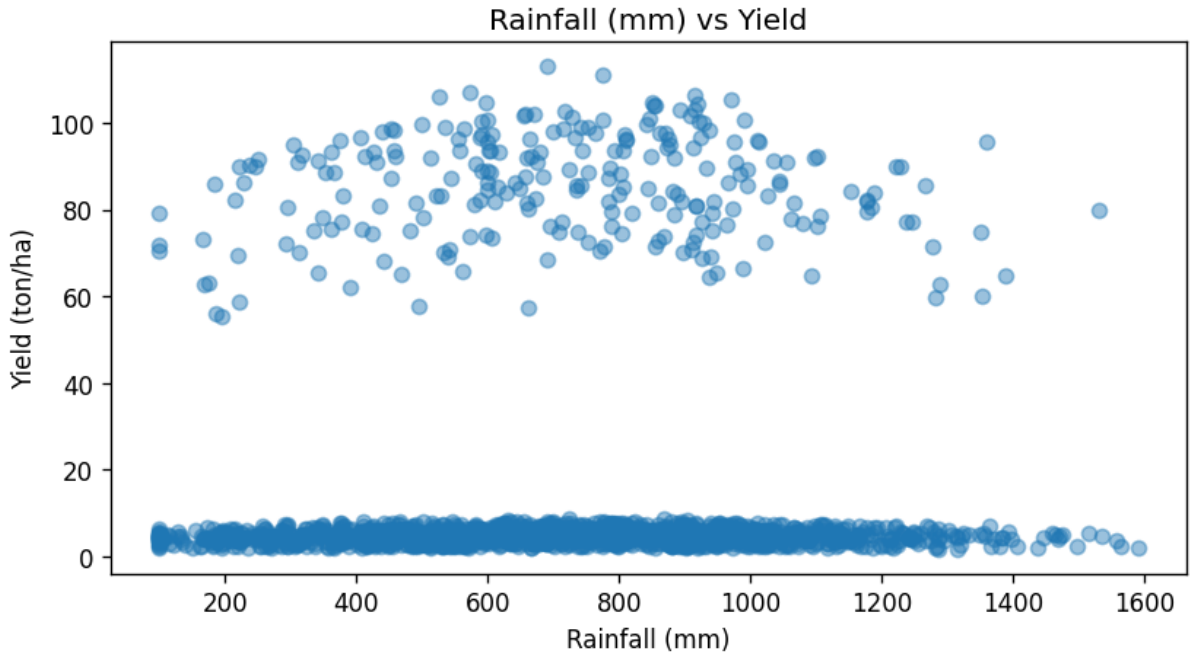
# 01 Yield Distribution



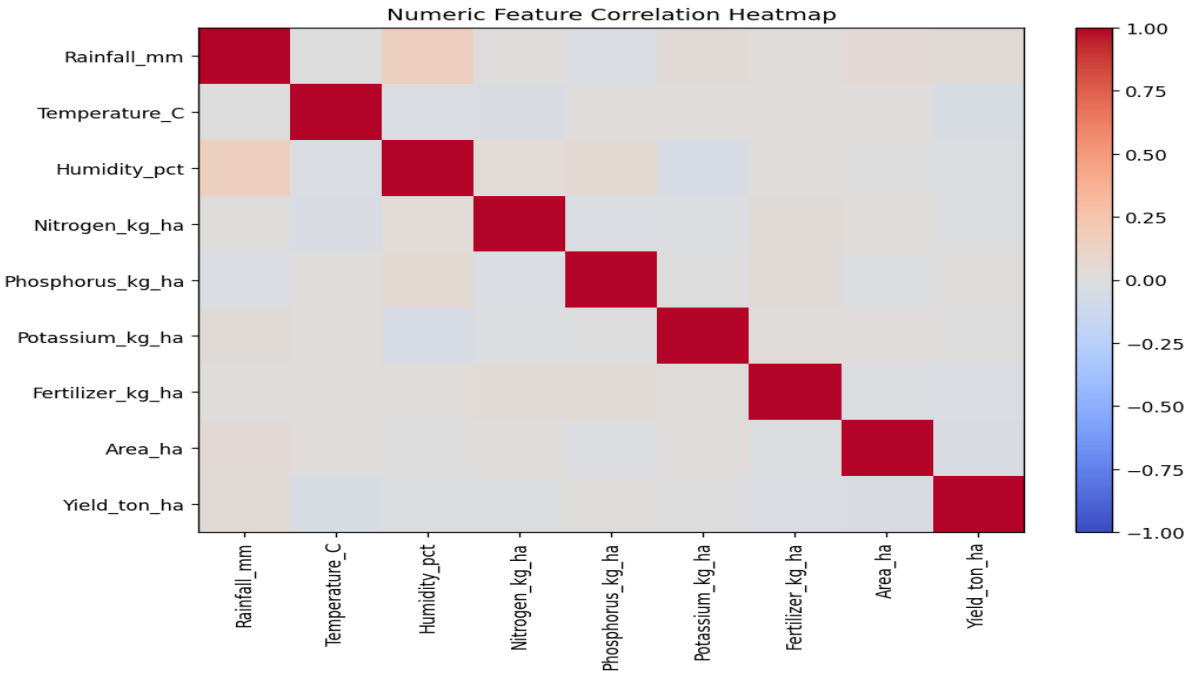
# 02 Mean Yield By Crop



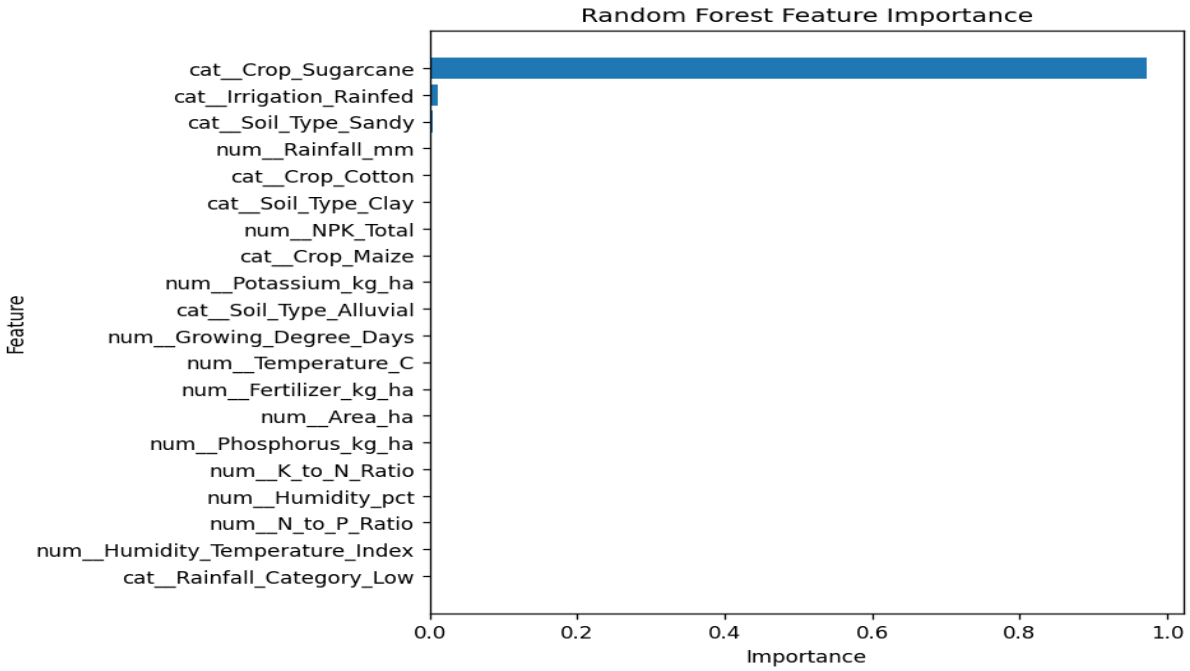
# 05 Rainfall Mm Vs Yield



# 15 Correlation Heatmap



# 17 Feature Importance



# 18 Residual Analysis

